

Dual Weight Pairing in the $GL(2, \mathbb{C})$ Quadratic Spinor Lagrangian

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Abstract

The $GL(2, \mathbb{C})$ extension [2] of the Tung–Jacobson quadratic spinor Lagrangian [1] replaces the traceless chiral connection by a $\mathfrak{gl}(2, \mathbb{C})$ -valued one, introducing a complex trace curvature $\mathcal{F} = d\mathcal{A}$ coupled to the gravitational spinor one-forms. The companion note [3] assigns to any spinor density a dilation weight W and a phase charge Δ , and shows that an invariant sesquilinear kinetic term with a weight-neutral volume element forces $W = 0$, so that no appended matter field of that type can be dilation-charged; the first exit it leaves open is a bilinear pairing between partners in dual weight representations. This paper observes that the quadratic spinor Lagrangian realises that exit algebraically in its gravitational sector and without appended matter: the one-forms α^A and β_A transform in dual $GL(2, \mathbb{C})$ representations, so their weights are equal and opposite and the pairing $\alpha^A \wedge \beta_A$, and with it the metric, is neutral in both channels. A conditional lemma is given: if an antilinear weight-neutral map from the conjugate spin space to the dual unprimed space relates the two one-forms, then dual-weight invariance forces $W(\alpha) = 0$, which is an incompatibility among assumptions rather than a dynamical statement. The result is therefore algebraic rather than dynamical: the gravitational potentials realise the dual-pairing exit from the sesquilinear matter no-go, while the survival of non-zero dilation weight under Lorentzian metric reality remains open. The trace curvature enters linearly through $B \wedge \mathcal{F}$ with $B = \alpha^A \wedge \beta_A$; its variation imposes $dB = 0$, but because B is composite this does not settle the local degree-of-freedom count.

Keywords: quadratic spinor Lagrangian, chiral gravity, spinor density, reality conditions.

1 Introduction

Two things are taken as given. The first is the quadratic spinor Lagrangian of Tung and Jacobson [1] and the $GL(2, \mathbb{C})$ extension of [2], recalled in Section 2. The second is the weight bookkeeping of [3], in which the trace of a $\mathfrak{gl}(2, \mathbb{C})$ connection splits into a real non-compact dilation potential $\mathfrak{a}$ and an imaginary compact phase potential $\mathfrak{b}$, a spinor density carries two real charges W and Δ formed from the sum and difference of its total determinant exponents, and complex conjugation preserves W while reversing Δ . The matter-coupling no-go of that note follows from the second of those facts: a sesquilinear pairing $\bar{\psi}\psi$ carries $W = 2W(\psi)$, so with a weight-neutral volume element an invariant kinetic term forces $W(\psi) = 0$. Its first exit is a bilinear pairing $\chi\psi$ with $W(\chi) = -W(\psi)$, in which χ is a partner in the dual weight representation rather than a complex conjugate.

The quadratic spinor Lagrangian is built on such a pairing. Its variables are two spinor-valued one-forms in dual spin spaces, the metric is their quadratic concomitant, and the combination appearing throughout the action is $\alpha^A \wedge \beta_A$. Section 3 records what that gives, which is a clean algebraic result. Section 4 states what it does not give, which is any conclusion

about the reality of the reconstructed metric, and formulates the calculation that would settle the matter. Section 5 treats the linear appearance of the trace curvature and is careful about what linearity does and does not establish. Section 6 collects the open problems in the order in which they would have to be answered.

2 Quadratic spinor Lagrangian and its $\mathrm{GL}(2, \mathbb{C})$ extension

Tung and Jacobson take a pair of two-component spinor-valued one-forms, or equivalently a single Dirac spinor one-form, and the chiral four-form

$$L_{\mathrm{TJ}} = D\alpha^A \wedge D\beta_A, \quad (1)$$

with D the exterior covariant derivative of an $\mathfrak{sl}(2, \mathbb{C})$ connection ω^A_B . The metric is the quadratic concomitant

$$g = \alpha^A \otimes \beta_A + \beta_A \otimes \alpha^A, \quad (2)$$

and in the two-component formulation (1) differs from the usual chiral action for general relativity by a total differential, while in the Dirac formulation the action is made real by adjoining the conjugate chiral sector [1]. Reality of the action and reality of the reconstructed metric are distinct conditions, and the distinction matters in Section 4. The spinor one-forms are therefore gravitational potentials rather than an appended matter multiplet, which is what makes the present exercise different in kind from that of [3].

The extension of [2] enlarges the internal symmetry by replacing D with the covariant derivative ∇ of a $\mathfrak{gl}(2, \mathbb{C})$ connection,

$$\hat{\omega}^A_B = \omega^A_B + \delta^A_B \mathcal{A}, \quad \mathcal{F} := d\mathcal{A}, \quad (3)$$

giving

$$L_{\mathrm{GL}(2, \mathbb{C})} = \nabla\alpha^A \wedge \nabla\beta_A = L_{\mathrm{TJ}} - B \wedge \mathcal{F} + d(B \wedge \mathcal{A}), \quad B := \alpha^A \wedge \beta_A, \quad (4)$$

so that the bulk difference between the two actions is a coupling of the trace curvature to the composite two-form B . Variation gives [2]

$$D(\alpha^{(A} \wedge \beta^{B)}) = 0, \quad (5)$$

$$D^2\beta_A - \beta_A \wedge \mathcal{F} = 0, \quad D^2\alpha^A + \alpha^A \wedge \mathcal{F} = 0, \quad (6)$$

$$dB = 0. \quad (7)$$

These include a spin-one field coupled to the chiral gravitational variables without being the Einstein–Maxwell equations in general: parts of the Ricci curvature are fixed by components of $\mathcal{F}$, and (7) differs from the source-free Maxwell equation, though plane-fronted solutions reduce to Einstein–Maxwell ones when the trace curvature is anti-self-dual [2].

Locally we write

$$\mathcal{A} = \mathfrak{a} + i\mathfrak{b}, \quad \mathcal{F} = d\mathfrak{a} + i d\mathfrak{b}, \quad (8)$$

with $\mathfrak{a}, \mathfrak{b}$ real and transforming as $\mathfrak{a} \mapsto \mathfrak{a} - d \ln |\xi|$, $\mathfrak{b} \mapsto \mathfrak{b} - d \arg \xi$ under a change of dyad by $\xi \in \mathrm{GL}(2, \mathbb{C})$, so that $\mathfrak{b}$ is a compact $\mathrm{U}(1)$ potential and $\mathfrak{a}$ a non-compact $\mathbb{R}^+$ potential [3].

Remark 2.1 (A caution about self-duality). The decomposition (8) should not be run together with the anti-self-duality of the special solutions. In Lorentzian signature $**H = -H$ for a real two-form H , and a nonzero anti-self-dual complex two-form is intrinsically complex, of the form $\mathcal{F} = H - i^*H$ with H real; requiring $\mathcal{F}$ to be pure imaginary and anti-self-dual at once forces $\mathcal{F} = 0$. The Einstein–Maxwell reduction of [2] therefore gives no support to any reading in which the surviving trace curvature is pure imaginary, and it is a standing caution against confusing the real and imaginary parts of $\mathcal{A}$ with the self-dual and anti-self-dual parts of $\mathcal{F}$.

3 Dual representations and the neutral pairing

We use the conventions of [3]: a spinor density with k, p upper and lower unprimed and l, q upper and lower primed indices and residual weights $(w, \bar{w})$ acquires $f^{2P} \bar{f}^{2Q}$ under $\xi^A_B = f^L{}^A_B$, with $P = w + \frac{1}{2}(k - p)$, $Q = \bar{w} + \frac{1}{2}(l - q)$, $W = P + Q$ and $\Delta = P - Q$, so that $\nabla\psi = (D + W\mathbf{a} + i\Delta\mathbf{b})\psi$; both charges are additive under tensor and wedge products and unchanged by raising or lowering indices with the ϵ 's, and conjugation preserves W and reverses Δ .

Proposition 3.1 (Duality fixes the pairing, whatever the magnitude). If α^A and β_A transform in dual $\mathrm{GL}(2, \mathbb{C})$ representations, in the sense that the pairing $\alpha^A\beta_A$ is a scalar of the enlarged group, then

$$W(\beta) = -W(\alpha) =: -\nu, \quad \Delta(\beta) = -\Delta(\alpha), \quad (9)$$

and consequently $W(B) = \Delta(B) = 0$ for $B = \alpha^A \wedge \beta_A$, and the metric (2) is neutral in both channels.

Proof. Additivity of W and Δ under products, applied to a pairing of vanishing total charge. $\square$

Corollary 3.2 (The sesquilinear no-go does not apply). The hypothesis of the no-go of [3] is a pairing $\bar{\psi}\psi$, which carries $W = 2W(\psi)$ because conjugation preserves W . The pairing here carries $W(\alpha) + W(\beta) = 0$ for any ν . The gravitational sector of (4) therefore admits fields of non-zero dilation weight without violating invariance, and it does so with no appended matter and no new field.

Duality therefore fixes the opposition of the two weights without fixing their common magnitude ν . That magnitude depends on the residual density weights and on how the $\mathrm{GL}(2, \mathbb{C})$ extension acts on the spinor-one-form decomposition.

Assumption 3.3 (Vanishing residual weights). If α^A and β_A carry vanishing residual weight, so that (3) acts on them through their single spin index alone, then $P(\alpha) = \frac{1}{2}$, $Q(\alpha) = 0$ and correspondingly for β , giving

$$W(\alpha) = \Delta(\alpha) = +\frac{1}{2}, \quad W(\beta) = \Delta(\beta) = -\frac{1}{2}. \quad (10)$$

Remark 3.4 (Why the residual weights are not obviously zero). Assumption 3.3 is an assumption and not a consequence of index position. In the quadratic spinor construction the one-forms arise from the co-frame after a primed spin frame has been chosen,

$$\alpha^A \sim \theta^{AA'} o_{A'}, \quad \beta^A \sim \theta^{AA'} \iota_{A'}, \quad (11)$$

so their determinant exponents inherit those of $\theta^{AA'}$, which the morphism normalisation of [3, §5] sets to $P = Q = 0$, together with those of the primed dyad, which the fixing of the primed frame has removed from view. If the normalisation $o_{A'}\iota^{A'} = 1$ is retained then the dyad elements carry opposite weights and Proposition 3.1 is recovered with $\nu = W(o)$; whether that equals $\frac{1}{2}$ depends on how the $\mathrm{GL}(2, \mathbb{C})$ extension acts on the decomposition, which is a question about [2] and not about index bookkeeping. Everything in this paper that matters holds for general ν ; Assumption 3.3 is quoted only where a numerical value is wanted.

There is a further reason not to lean on the magnitude. W is defined relative to a normalisation of the trace connection, and under $\mathcal{A} \mapsto c\mathcal{A}$ with $\nu \mapsto \nu/c$ the product $\nu\mathcal{A}$ appearing in the covariant derivative is unchanged. Until some other sector fixes that normalisation, the invariant content is whether ν vanishes, whether the two weights are opposite, and the ratios of independently occurring charges; the value $\frac{1}{2}$ is convention-dependent.

4 Reality

The reality condition of the source theory is imposed on the reconstructed metric. In the Dirac formulation the action is made real by adjoining the conjugate chiral sector, and that action is real for general complex spinor one-forms; reality of the metric remains the independent condition $\Psi \otimes_S \gamma_5 \Psi = 0$ [1], so a real action does not deliver a real metric. In neither formulation is the condition an identification of the two one-forms with one another, and it could not be: complex conjugation carries an unprimed spinor to a primed one, so producing a lower unprimed β_A from α^A requires an antilinear map from the conjugate spin space to the dual unprimed space, which Lorentzian spin geometry does not supply canonically without further structure. The following is therefore stated conditionally, and its content is narrower than it may first appear.

Lemma 4.1 (Conditional). Suppose there exists an antilinear map $J : \bar{S} \rightarrow S^*$, weight-neutral in the sense that $W(J\psi) = W(\bar{\psi})$ and $\Delta(J\psi) = \Delta(\bar{\psi})$, with $\beta = J\bar{\alpha}$. Then $W(\beta) = +W(\alpha)$, since conjugation preserves W , while Proposition 3.1 requires $W(\beta) = -W(\alpha)$; hence $W(\alpha) = 0$. The phase charges are consistent for any value, since conjugation reverses Δ and duality requires the same reversal.

Remark 4.2 (What the lemma is a statement about). A representation charge is not a dynamical variable that a condition can set to zero on shell. If α transforms in the fundamental with $W(\alpha) = \nu$, it does not acquire $W = 0$ when the metric is made real. Lemma 4.1 is an incompatibility, and the alternatives it leaves are three:

1. no such weight-neutral J exists for representations with $\nu \neq 0$;
2. the residual weights are not those of Assumption 3.3, and the correct assignment has $\nu = 0$, which is a different representation from the one used to state the assumption;
3. the structure group acting on the potentials is reduced, so that the transformation whose exponent W counts no longer acts.

Which of the three obtains is not decided by the weight bookkeeping. It is decided by the reality condition of the source theory, which the bookkeeping has not yet been applied to.

Remark 4.3 (Decoupling is not vanishing). Even where $W = 0$ for the potentials, what follows is that $\mathfrak{a}$ does not appear in $\nabla\alpha$ and $\nabla\beta$. It does not follow that $\mathfrak{a} = 0$ or that $d\mathfrak{a} = 0$; a connection is removable by a gauge choice only where its curvature vanishes, and (4) retains $\mathcal{F}$ through the coupling to B regardless of how the potentials transform. Concluding $\mathcal{A} = i\mathfrak{b}$ would require an explicit reduction of the connection, a metricity condition, a field equation giving $d\mathfrak{a} = 0$ followed by a local gauge choice, or a reality condition imposed directly on the connection rather than on the metric. None of these is established here. The distinction between absence of a direct dilation vertex and absence of a real trace curvature is the one the separation of this paper from [3] exists to preserve.

The calculation these remarks point to is the following, and it is the main open problem of this paper.

Open problem 4.4 (The metric-reality condition in weight terms). Determine how the conditions

$$\text{Im } g = 0, \quad \det g \neq 0, \quad \text{sig}(g) = (-, +, +, +), \quad (12)$$

for g the concomitant (2), equivalently $\Psi \otimes_S \gamma_5 \Psi = 0$ together with non-degeneracy in the Dirac formulation [1], constrain the determinant exponents of α^A and β_A and the real trace connection $\mathfrak{a}$. Specifically: does a real, non-degenerate metric of Lorentzian signature admit $\nu \neq 0$, and if it does not, do split or Euclidean signatures admit it?

Proposition 3.1 says that the pairing is invariant for any ν , so nothing in the invariance ledger obstructs $\nu \neq 0$. Lemma 4.1 says that one particular class of relations between the two one-forms would obstruct it. Problem 4.4 asks whether the relation actually imposed does. The honest summary is that the dual-pairing exit from the matter no-go is available algebraically and that its survival under Lorentzian reality is undecided.

5 Trace entering linearly

Independently of reality, the form of (4) constrains what the trace sector can be.

Proposition 5.1. $\mathcal{F}$ appears in (4) linearly and wedged with B rather than with ${}^*\mathcal{F}$. Variation with respect to $\mathcal{A}$ therefore yields $dB = 0$, which is (7), and no equation of Maxwell form $d{}^*\mathcal{F} = 0$. In consequence $\mathcal{A}$ has no conventional Maxwell kinetic term, and the action supplies no Maxwell-type momentum independent of the gravitational one-forms: the momentum conjugate to the spatial components of $\mathcal{A}$, where defined, is constrained to be a component of the composite B .

Proof. $-B \wedge d\mathcal{A}$ has variation $-dB \wedge \delta\mathcal{A}$ up to an exact form. $\square$

This is the structural reason behind the observation of [2] that (7) differs from the source-free Maxwell equation. It does not establish that the trace sector carries no propagating mode. B is not an independent BF two-form; it is a composite of the dynamical one-forms, which carry derivative terms of their own, and a field appearing without an $\mathcal{F} \wedge {}^*\mathcal{F}$ term may still participate in propagating mixed modes once the constraints are solved. The source paper describes the extension as containing a spin-one field, and displacing that description requires a constraint analysis and a local degree-of-freedom count, neither of which is performed here. A constrained momentum of this kind is a source of primary constraints and of kinetic mixing rather than of an independent Maxwell electric field, so what is needed is a Hamiltonian constraint count; until it is done, what can be said is the content of Proposition 5.1 and no more, and the question of whether either trace channel propagates is not answered by the extended quadratic spinor action either way.

Remark 5.2 (What a quadratic term would have to be). Supplying a conventional kinetic term means adding $\mathcal{F} \wedge {}^*\mathcal{F}$, which is metric-dependent and therefore couples the trace sector to (2). The topological alternative $\mathcal{F} \wedge \mathcal{F} = d(\mathcal{A} \wedge \mathcal{F})$ is exact for an abelian connection and supplies no local field equation and no propagating mode. A constrained BF construction of the kind used in [6, 8] may generate effective dynamics after auxiliary fields are eliminated, and that is a calculation rather than an observation. The three cases differ in what they cost. The metric-dependent term changes the bulk dynamics and breaks the equivalence of (1) to the chiral action, which is what makes the one-forms gravitational potentials rather than matter. The topological term changes only boundary and topological data, since a relation holding up to a total differential is not destroyed by adding another one. The constrained construction may or may not change the effective bulk theory, depending on the elimination of the auxiliary fields.

Remark 5.3 (On the coefficient). Both B and $\mathcal{F}$ are invariant under the abelian trace transformation, so $B \wedge \mathcal{F}$ does not transform and cannot be what cancels the inhomogeneous change of L_{TJ} under the enlarged group; that cancellation is supplied by the boundary term, which shifts by $-d(B \wedge d\lambda)$ under $\mathcal{A} \mapsto \mathcal{A} - d\lambda$. The value unity in (4) is therefore fixed by the covariant-square origin of the action, by the requirement that it arise exactly as $\nabla\alpha^A \wedge \nabla\beta_A$, and not by invariance of the coupling term on its own. A general $L_\kappa = L_{\text{TJ}} - \kappa B \wedge \mathcal{F}$ accordingly defines a deformation of the quadratic spinor extension rather than another

expansion of the same covariant derivative, and whether $\kappa \neq 0$ is physically distinct depends on the normalisation of $\mathcal{A}$, which a rescaling can absorb, and on whatever further trace-sector terms are present to fix it.

6 Open problems

1. Problem 4.4, the weight content of the metric-reality condition. Everything else waits on it, since it decides whether $\nu \neq 0$ is available in Lorentzian signature at all, and since it is the point at which the trichotomy of Remark 4.2 is resolved rather than listed.
2. Given an admissible reality structure with $\nu \neq 0$, the kinetic bilinear and the stress-energy of the trace sector, and a constraint analysis and local degree-of-freedom count for (4), with and without the quadratic term of Remark 5.2, together with positivity or its failure.
3. The relation between the composite two-form $B = \alpha^A \wedge \beta_A$ of Proposition 5.1 and the multiplier trace of the constrained BF formulations [8], whose discarded scalar is discussed in [3, §7]. Both formulations share a $\mathrm{GL}(2, \mathbb{C})$ origin and in both the trace enters linearly, which makes an identification plausible and does not make it true.
4. Whether the removal of the dilation weight, in whichever of the three forms of Remark 4.2 the answer to Problem 4.4 takes, is a feature of the particular reality condition or of Lorentzian signature itself. This is the question the two papers together were written to reach.

What is claimed here is one algebraic statement and its consequence: α^A and β_A transform in dual $\mathrm{GL}(2, \mathbb{C})$ representations, so their weights are equal and opposite and the pairing that builds the metric is neutral in both channels; the quadratic spinor gravitational potentials therefore realise the dual-pairing exit from the sesquilinear matter no-go of [3] without any appended field. Whether that exit survives Lorentzian reality, and whether the trace connection carries any local dynamical mode, remain unresolved, and neither is settled by the bookkeeping that establishes the first statement.

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